

Ordinal Regression

$$\vec{x}_i = (\underbrace{x_{i1}, x_{i2}, \dots, x_{ip}}_{\text{length} = p}) \quad i=1, 2, \dots, n \quad (n \text{ observations})$$

$$y_i = \{1, 2, \dots, K\} \quad \text{ordinal variable}$$

* hypothesis function:

$$h_{\theta_i}(\vec{x}) = \theta_i - \vec{w} \cdot \vec{x}, \quad \begin{cases} \vec{w} = (w_1, w_2, \dots, w_p) \\ \theta_1 < \theta_2 < \dots < \theta_{K-1} \end{cases}$$

$$* \text{ model: } \Pr(y \leq i | \vec{x}) = \sigma(\theta_i - \vec{w} \cdot \vec{x})$$

$$\textcircled{1} \text{ logistic function: } \sigma(\theta_i - \vec{w} \cdot \vec{x}) = \frac{1}{1 + \underbrace{e^{-(\theta_i - \vec{w} \cdot \vec{x})}}_{e^{-\theta_i x}}}$$

$$\textcircled{2} \text{ probit function}$$

$$\textcircled{3} \text{ exponential function: } \sigma(\theta_i - \vec{w} \cdot \vec{x}) = \frac{1}{e^{(\theta_i - \vec{w} \cdot \vec{x})}}$$

App: movie rating on a scale 1 of 5